

JC-Computation: Formal Theory

Author: James Chapman xhecarpenxer@gmail.com

Part I — Confluence of NF (Formal Proof)

1. Setup

We work with the Abstract Rewriting System (ARS) $(H, \rightarrow_R)$ where:

- H is the set of all causal histories $H = (E, <, \lambda)$
- $\rightarrow_R$ is the single-step reduction relation defined by the NF operator's four phases

We write $\rightarrow_{R^*}$ for the reflexive-transitive closure (many reduction steps).

A history H^* is in **normal form** if no rule in R applies to it, i.e., there is no H' with $H \rightarrow_R H'$.

2. Definitions

Definition (Confluence). A rewriting system $(H, \rightarrow_R)$ is *confluent* if for all (H, H_1, H_2) :

$$\begin{aligned} H \rightarrow_{R^*} H_1 \quad \text{and} \quad H \rightarrow_{R^*} H_2 \\ \implies \\ \exists H^* : H_1 \rightarrow_{R^*} H^* \quad \text{and} \quad H_2 \rightarrow_{R^*} H^* \end{aligned}$$

Definition (Local Confluence / Weak Church-Rosser). The system is *locally confluent* if for all (H, H_1, H_2) :

$$\begin{aligned} H \rightarrow_R H_1 \quad \text{and} \quad H \rightarrow_R H_2 \\ \implies \\ \exists H^* : H_1 \rightarrow_{R^*} H^* \quad \text{and} \quad H_2 \rightarrow_{R^*} H^* \end{aligned}$$

Definition (Termination / Strong Normalization). The system is *terminating* if there is no infinite reduction sequence $H_0 \rightarrow_R H_1 \rightarrow_R H_2 \rightarrow_R \dots$

Theorem (Newman's Lemma). If a rewriting system is both *terminating* and *locally confluent*, then it is *confluent*.

We prove termination and local confluence separately, then apply Newman's Lemma.

3. Termination Proof

Theorem 3.1 (NF Termination). The reduction system $(H, \rightarrow_R)$ is terminating.

Proof.

Define a complexity measure $\Phi : H \rightarrow \mathbb{N}^3$ (lexicographically ordered):

$$\Phi(H) = (|E|, \text{entropy}(H), \text{disorder}(H))$$

where:

- $|E|$ = number of events in H
- $\text{entropy}(H)$ = number of distinct causal cone hash classes (decreases on merge)
- $\text{disorder}(H)$ = number of independent event pairs not yet canonically ordered

We show each reduction phase strictly decreases Φ under $<_{\text{lex}}$:

Phase A (Causal Closure): Adds events, so $|E|$ can only increase or stay equal. However, Phase A only fires when an ancestor is missing — a condition that after closure is resolved. Phase A is therefore idempotent after first application and does not fire again. Φ unchanged after stabilization. ✓

Phase C1 (Cone merging): Two events (e_1, e_2) with $\text{coneHash}(e_1) = \text{coneHash}(e_2)$ are merged into one. Thus $|E|$ strictly decreases by at least 1:

$$\begin{aligned} |E'| &= |E| - 1 < |E| \\ \Rightarrow \Phi(H') &<_{\text{lex}} \Phi(H) \quad \checkmark \end{aligned}$$

Phase C2 (Linear chain contraction): Removes an intermediate event (b) in $(a \rightarrow b \rightarrow c)$ when (b) has no semantic effect. Thus $|E|$ strictly decreases:

$$|E'| = |E| - 1 < |E| \quad \checkmark$$

Phase C3 (No-op elimination): Removes a no-op event. $|E|$ strictly decreases. ✓

Phase B (Canonical ordering): Does not change $|E|$. However it strictly decreases $\text{disorder}(H)$ because it places all independent pairs in canonical order — a one-pass idempotent operation. Once applied, no pair is disordered:

$$|E'| = |E|, \text{entropy}(H') = \text{entropy}(H), \text{disorder}(H') = 0 < \text{disorder}(H) \quad \checkmark$$

(unless $\text{disorder}(H) = 0$ already, in which case Phase B is a no-op and Φ is unchanged — no infinite loop)

Phase D (Hash stabilization): Purely recomputes identifiers. Does not change the graph structure. Idempotent. Φ unchanged. ✓

Conclusion: Every non-trivial reduction step either decreases $|E|$ (Phases C1, C2, C3) or decreases disorder with $|E|$ fixed (Phase B), or is a no-op (Phases A, D after stabilization). Since $|E|$ is a natural number bounded below by 0, and disorder is bounded below by 0, and the lexicographic order on $\mathbb{N}^3$ is well-founded, no infinite reduction sequence exists.

$\therefore (H, \rightarrow_R)$ is **terminating**. $\square$

4. Local Confluence Proof

Theorem 4.1 (NF Local Confluence). The reduction system $(H, \rightarrow_R)$ is locally confluent.

Proof.

We must show that for any H with two applicable rules (r_1, r_2) producing (H_1, H_2) , there exists H^* reachable from both.

The rules are drawn from: $\{A, B, C1, C2, C3, D\}$. We check all critical pairs (cases where two rules apply to the same or overlapping substructures):

Case (C1, C1): Two independent cone merges

Suppose $\text{coneHash}(a) = \text{coneHash}(b)$ and $\text{coneHash}(c) = \text{coneHash}(d)$, where $\{a, b\} \cap \{c, d\} = \emptyset$.

- r_1 merges $(a, b) \rightarrow H_1$
- r_2 merges $(c, d) \rightarrow H_2$

In H_1 : rule for $\{c, d\}$ still applies (untouched). Apply it $\rightarrow H^*$ In H_2 : rule for $\{a, b\}$ still applies (untouched). Apply it $\rightarrow H^*$

Both reach the same H^* because the merges act on disjoint subgraphs and cone hashes are content-addressed. Confluent. ✓

Case (C1, C1): Overlapping cone merges (shared ancestor)

Suppose $\text{coneHash}(a) = \text{coneHash}(b)$ and $\text{coneHash}(b) = \text{coneHash}(c)$.

This means $\text{coneHash}(a) = \text{coneHash}(b) = \text{coneHash}(c)$ — all three are in the same equivalence class.

- r_1 merges $(a, b) \rightarrow H_1$ (b eliminated, edges redirected to a)
- r_2 merges $(b, c) \rightarrow H_2$ (c eliminated, edges redirected to b)

In H_1 : c still exists with $\text{coneHash}(a) = \text{coneHash}(c)$. Apply $C1 \rightarrow \text{merge } (a, c) \rightarrow H^*$ with single representative. In H_2 : a still exists with $\text{coneHash}(a) = \text{coneHash}(b_representative)$. Apply $C1 \rightarrow H^*$.

Both reach the same canonical representative of the equivalence class. ✓

Case (C2, C2): Two independent chain contractions

Linear chains $(a \rightarrow b \rightarrow c)$ and $(p \rightarrow q \rightarrow r)$ with $\{a, b, c\} \cap \{p, q, r\} = \emptyset$.

- r_1 contracts $b \rightarrow H_1$
- r_2 contracts $q \rightarrow H_2$

Contractions act on disjoint subgraphs. Apply the other contraction to reach H^* .

Confluent. ✓

Case (C2, C2): Nested chain $(a \rightarrow b \rightarrow c \rightarrow d)$

- r_1 contracts b (middle of $(a \rightarrow b \rightarrow c)$) $\rightarrow (a \rightarrow c \rightarrow d) \rightarrow H_1$
- r_2 contracts c (middle of $(b \rightarrow c \rightarrow d)$) $\rightarrow (a \rightarrow b \rightarrow d) \rightarrow H_2$

In H_1 ($(a \rightarrow c \rightarrow d)$): c now has single parent a , single child d , no payload $\rightarrow \text{contract } c \rightarrow (a \rightarrow d) = H^*$. In H_2 ($(a \rightarrow b \rightarrow d)$): b now has single parent a , single child d , no payload $\rightarrow \text{contract } b \rightarrow (a \rightarrow d) = H^*$.

Both reach $(a \rightarrow d)$. ✓

Case (C1, C2): Cone merge and chain contraction

If the chain being contracted $(a \rightarrow b \rightarrow c)$ has b with a cone isomorphic to some other event, and simultaneously $C1$ wants to merge b with that event:

After $C1$ merges b with its isomorphic copy: the resulting chain may or may not still be contractible. If it is, $C2$ fires. If it is not (merged node now has branching), $C2$ does not fire.

Either way a unique result is reached because the graph is finite and both operations decrease (Φ) . ✓

Case (B, anything): Canonical ordering vs structural rules

Phase B only adds ordering edges between independent events. It does not remove events or change payloads. All of C1, C2, C3 operate on event identities and payloads, not ordering edges among independent events. Therefore B and C-rules commute: apply either first, then the other, reach (H^*) . ✓

Case (C3, anything): No-op elimination

No-op events by definition have $(\lambda(e) = \text{noop})$ and contribute nothing to cones or chain semantics. Removing them does not affect applicability of any other rule. Any order of removal reaches the same (H^*) . ✓

Conclusion: All critical pairs join. $\therefore (H, \rightarrow_R)$ is **locally confluent**. □

5. Main Confluence Theorem

Theorem 5.1 (NF Confluence). The JC normal form reduction system is confluent.

Proof. By Theorem 3.1 (Termination) and Theorem 4.1 (Local Confluence), Newman's Lemma applies directly:

Terminating + Locally Confluent $\Rightarrow$ Confluent

$\therefore$ For any history (H) and any two reduction sequences reaching (H_1) and (H_2) , there exists a unique (H^*) (up to isomorphism) reachable from both. □

Corollary 5.2 (Uniqueness of Normal Form). Every history (H) has a unique normal form $(nf(H))$.

Proof. By confluence, all maximal reduction sequences from (H) reach the same result. By termination, all reduction sequences are finite and thus maximal. □

6. Convergence Theorem (State Invariance)

Theorem 6.1. Let $(\sigma : H \rightarrow S)$ be the semantic functor. If $(H_1 \approx H_2)$ (same causal equivalence class), then $(\sigma(nf(H_1)) = \sigma(nf(H_2)))$.

Proof. By Corollary 5.2, $\boxed{\text{nf}(H_1) = \text{nf}(H_2)}$ (unique normal form for equivalent histories).
Therefore $\boxed{\sigma(\text{nf}(H_1)) = \sigma(\text{nf}(H_2))}$. $\square$

This is the central theorem: **state is a property of causal history, not of execution order.**

Part II — Category-Theoretic Model

7. The Category JC

Define the category **JC** as follows:

Objects: Causal histories $\boxed{H = (E, <, \lambda)}$ in normal form.

Morphisms: History extensions $\boxed{f : H \rightarrow H'}$ where:

- $\boxed{H \subseteq H'}$ (H' extends H)
- $\boxed{f}$ preserves the causal relation: $\boxed{e_1 <_H e_2 \Rightarrow e_1 <_{\{H'\}} e_2}$
- $\boxed{f}$ is admissible: $\boxed{A(H, e)}$ holds for each new event $\boxed{e}$

Composition: Given $\boxed{f : H \rightarrow H'}$ and $\boxed{g : H' \rightarrow H''}$, composition $\boxed{g \circ f : H \rightarrow H''}$ is the combined extension.

Identity: $\boxed{\text{id}_H : H \rightarrow H}$ is the empty extension (add no events).

Theorem 7.1. **JC** is a well-defined category.

Proof.

- *Identity laws:* $\boxed{\text{id}_{\{H'\}} \circ f = f}$ and $\boxed{f \circ \text{id}_H = f}$ trivially hold since identity extensions add nothing.
 - *Associativity:* $\boxed{(h \circ g) \circ f = h \circ (g \circ f)}$ holds because union of event sets is associative. $\square$
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8. The Semilattice Subcategory

The merge operation $\boxed{\oplus : H \times H \rightarrow H}$ defined by:

$$H_a \oplus H_b = \text{nf}(H_a \cup H_b)$$

equips the objects of **JC** with a join-semilattice structure under the partial order $\boxed{H \sqsubseteq H'}$ iff $\boxed{H \subseteq H'}$.

Theorem 8.1. $\boxed{(\text{Obj}(\text{JC}), \sqsubseteq, \oplus)}$ is a join-semilattice.

Proof.

- *Commutativity*: $(H_a \oplus H_b = \text{nf}(H_a \cup H_b) = \text{nf}(H_b \cup H_a) = H_b \oplus H_a)$ (set union commutes).
 - *Associativity*: $((H_a \oplus H_b) \oplus H_c = \text{nf}(\text{nf}(H_a \cup H_b) \cup H_c) = \text{nf}(H_a \cup H_b \cup H_c))$ by confluence of nf (adding already-normal events is idempotent). Associativity follows.
 - *Idempotency*: $(H \oplus H = \text{nf}(H \cup H) = \text{nf}(H) = H)$ (H already in normal form). $\square$
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9. The State Functor

Definition. The *state functor* $(\Sigma : \mathbf{JC} \rightarrow \mathbf{Set})$ is defined by:

$$\begin{aligned}\Sigma(H) &= \sigma(H) && \text{(on objects)} \\ \Sigma(f : H \rightarrow H') &= \sigma(H') && \text{(on morphisms – state after extension)}\end{aligned}$$

Theorem 9.1 (Functoriality of Σ). (Σ) is a functor.

Proof.

- *Preserves identity*: $(\Sigma(\text{id}_H) = \sigma(H) = \text{id}_{\{\sigma(H)\}})$ $\checkmark$
 - *Preserves composition*: $(\Sigma(g \circ f) = \sigma(H'') = \Sigma(g) \circ \Sigma(f))$ (state after composed extension equals state after extension). $\checkmark \square$
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10. The Quotient Functor

Define the equivalence relation $(\sim)$ on morphisms: $(f \sim g)$ iff $(\text{nf}(H_f) = \text{nf}(H_g))$ (same normal form history reached).

The quotient category $(\mathbf{JC}/\sim)$ has:

- Same objects as $\mathbf{JC}$
- Morphisms are equivalence classes $([f])$ under $(\sim)$

Definition. The *quotient functor* $(Q : \mathbf{JC} \rightarrow \mathbf{JC}/\sim)$ sends each morphism to its equivalence class:

$$\begin{aligned}Q(H) &= H \\ Q(f) &= [f]\end{aligned}$$

Theorem 10.1. There exists a unique factorization:

$$\Sigma = \bar{\Sigma} \circ Q$$

where $(\bar{\Sigma} : JC/\sim \rightarrow Set)$ is a well-defined functor on the quotient.

Proof. By confluence (Theorem 5.1), $(\bar{\Sigma})$ is constant on equivalence classes of morphisms (two morphisms reaching the same nf reach the same state). By the universal property of quotient categories, $(\bar{\Sigma})$ factors uniquely through (Q) . $\square$

This is the category-theoretic form of the Representation Theorem: **state is a functor on the quotient category of causal histories.**

11. Natural Transformations and Protocol Morphisms

A *protocol morphism* between two JC systems $((D_1, \Sigma_1))$ and $((D_2, \Sigma_2))$ is a natural transformation $(\eta : \Sigma_1 \Rightarrow \Sigma_2)$ — a family of maps $(\eta_H : \Sigma_1(H) \rightarrow \Sigma_2(H))$ commuting with all extensions.

This gives a 2-category structure where:

- Objects: JC-Computation systems
- 1-morphisms: history-preserving maps
- 2-morphisms: natural transformations (protocol refinements)

Corollary 11.1. Database schema migrations, consensus protocol upgrades, and CRDT type changes are all natural transformations in this 2-category.

Part III — Complexity Summary

Operation	Time Complexity	Space Complexity
Event ingestion	$O(\log n)$ amortized	$O(1)$ per event
Causal closure	$O(n)$ incremental	$O(n)$
Concurrency canonicalization	$O(n \log n)$	$O(n)$
Cone hash computation	$O(\text{depth})$ per node	$O(n)$
Cone merge detection	$O(n)$ with hash index	$O(n)$

Operation	Time Complexity	Space Complexity
Linear chain contraction	$O(n)$	$O(n)$
Full NF pass	$O(n \log n)$	$O(n)$
Distributed merge	$O(n \log n)$	$O(n)$
State derivation	$O(n)$	$O(n)$

Full NF convergence: $O(n^2 \log n)$ worst case, $O(n \log n)$ amortized in structured systems.

End of formal theory.